

FORMULATION OF A SELF-EMULSIFYING LIPID FORMULATION OF CURCUMIN

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INTRODUCTION

The aim of this study was to develop a Self Emulsifying Lipid Formulation (SELF) of curcumin with the sound formulation protocol developed by Gattefossé following recommendations from the LFCS Consortium. The fate of a BCS class II drug in a SELF depends on the ability of the formulation components to keep the drug in solution during the initial dispersion and/or digestion processes. These systems require a suitable selection of excipients and an appropriate combination of oils, water-soluble surfactants, water-insoluble surfactants, and/or hydrophilic cosolvents based on their intended function: some excipients are needed to dissolve the drug within the dosage form; others to generate the metabolites that maintain drug solubility in the gastro-intestinal colloidal phases.

EXPERIMENTAL METHODS

Curcumin was chosen as a poorly-water soluble model compound with logP=3. Liquid and solid excipients tested in this study are presented in Table 1. These excipients are classified in four types: water-soluble surfactants, water-insoluble surfactants a.k.a. co-surfactants, oils, and hydrophilic cosolvents (Table 1).

• Solubility of drugs in liquid excipients

Solubility of curcumin in liquid excipients is assayed by HPLC. In brief, the drug is added in excess to 60 mL amber borosilicate glass bottle containing 10 g lipid-based excipient. Bottle are allowed to equilibrate at 37°C under magnetic stirring with periodic vortex mixing to ensure that undissolved drug particles are homogeneously suspended in the lipid slurry. At intervals, bottles are sampled and these aliquots are centrifuged (Universal centrifuge 320R) at 2800g and 37°C for 30 min [2]. Samples are separated into a solid pellet phase and a particle-free supernatant. The supernatant is sampled and diluted with methanol for further HPLC analysis. The equilibrium solubility is considered as reached when the difference between two consecutive values is less than 5%. For most of the excipients, the equilibrium is reached in less than 3 days.

• Solubility of drugs in solid excipients

For solid lipid-based excipients, the solubility of curcumin is assayed by differential scanning calorimetry (DSC). According to the Raoult's law, the variation of melting enthalpy is related to the concentration of drug dissolved within the excipient. Samples are prepared by melting the excipient and dispersing a specified amount of drug under stirring. These samples are left at 50°C to equilibrate overnight, vortexed, then aliquoted in aluminum pans (2.5-5 mg), and left at 25°C to solidify / crystallize for 24 hours before DSC analysis (Perkin-Elmer Pyris Diamond). Thermal analysis is carried out between -20 and 120°C with a heating rate of 3°C.min⁻¹. The drug solubility is observed when there is a change of slope in the curve fitting the evolution of melting enthalpy of the excipient as a function of the drug concentration in the sample [3]. Solubility is then confirmed by polarized light hot stage microscopy (PL-HSM, Zeiss AxioScope A11).

• Dispersion of selected formulation in water

A dispersion test is performed on selected mixtures in a USP dissolution bath. One gram of mixture is introduced in 200 mL of purified water at 37°C under agitation with a paddle at 100 rpm. Performance criteria are: ease of emulsification, homogeneity, fineness of the dispersion, and absence of curcumin precipitation. The fineness of the dispersion is assayed by dynamic light scattering (DLS, Particle Sizing Systems Nicomp). The precipitation of curcumin is checked by polarized light microscopy (PLM).

• Preparation of media and enzyme suspension

The lipolysis buffer is prepared by adding Tris (0.474 g/L), CaCl₂ (0.208 g/L) and NaCl (8.810 g/L) in Milli-Q water. The pH is adjusted at 6.5 with either HCl 1M or NaOH 0.6 M. The lipolysis medium is prepared by adding L-α-phosphatidylcholine (0.576 g/L) and NaTDC (0.576 g/L) in the lipolysis buffer. The medium is stirred overnight to allow the complete solubilisation of lecithin.

The pancreatin solution is prepared by adding 1 g of pancreatin powder in 5 mL of lipolysis buffer. After 10 minutes of magnetic stirring the solution is centrifuged at 2800g and 5°C for 10 min. The supernatant is sampled to be used in the lipolysis test [2].

• In vitro lipolysis test

The experimental setup consists in a pH stat apparatus (Metrohm®), comprising a Titrand 802 propeller stirrer/804 Ti Stand combination, a glass pH electrode (Inlitrade) and one 800 Dosino dosing units coupled to 5 mL autoburette. The apparatus is connected to a PC and operated using Tiamo 2.0 software. The equipment is thermostated at 37°C and filled with lipolysis medium, to mimic fasted state in the small intestine. During the digestion of lipid-based formulations by pancreatic lipases, pH is maintained constant by the addition of NaOH 0.1N.

Before the lipolysis test, the specific activity of pancreatic lipases is checked on tributyrine (model substrate). The substrate must be in excess to have a reaction rate directly proportional to the enzyme concentration. Five hundred microliters of tributyrine and 1.5 µL of pancreatin suspension are introduced in 28 mL of lipolysis medium (without lecithin). The specific activity of pancreatic lipases should be higher than 900 U/mg.

One gram of lipid-based formulations in hard gelatine capsules (Licaps, Capsugel), is added to 36 mL of lipolysis medium at 37°C. After 10 minutes (dispersion / emulsification time), 4 mL of pancreatin solution is added. The duration of the lipolysis is 1 hour. Aliquots of 1 mL are sampled at various time points to determine the solubility of the drug in the micellar phase: t=5 min (during the dispersion time), t=0 min (before the addition of the pancreatic solution), t=5 min, t=15 min, t=30 min and t=60 min. Five microliters of the inhibitor solution (1M: 100 mg of 4-bromophenylboronic acid in 0.5 mL of methanol) is added immediately to each sample. They are then centrifuged at 21 000g and 37°C for 30 minutes. The supernatant is sampled and diluted with methanol for further HPLC analysis.

Placebo formulations are also tested in order to determine the solubility of the drug in the lipolysis medium containing the digested lipid formulation. In this case only one sampling is performed at t=60 min. The micellar phase is obtained at the end of lipolysis experiment after centrifugation at 21 000g and 37°C for 30 minutes of aliquots of the pH-stat medium. The supernatant is sampled and drug is added in excess. Mixtures are let at 37°C under agitation for 4 hours. Aliquots are then centrifuged as described above and diluted with methanol for HPLC analysis.

RESULTS AND DISCUSSION

• Solubility determination and formulation selection

Excipients exhibiting the higher solvent capacity were water-soluble surfactants and Transcutol®, the hydrophilic co-solvent (Table 1). All these excipients were able to dissolve more than 30 mg/mL of curcumin, the target dose for this study. We decided then to select as prototype formulations all mixtures of curcumin with water-soluble surfactants which are able to self-emulsify in contact with water and leading to clear and transparent dispersions.

• Dispersion test

All prototypes were able to form transparent and homogeneous dispersions in contact with water and to keep curcumin in solution (Table 2). A swelling of the colloidal phase is observed for all prototype formulations after addition of the compound to the placebo. No precipitation was observed by PLM. Classic dispersion test was not able to discriminate the four prototype formulations.

Table 1: Solubility of curcumin in lipid-based excipients at 37°C.

Functionality	Chemical name	Commercial name	Solubility (mg/mL)
Surfactants (water-soluble)	Polyethylene-20 sorbitan oleate	Tween® 80	88.4
	PEG-32 stearate	Gelucire® 48/16	70.0
	Caprylo caproyl polyoxyl-8 glycerides	Labrasol® ALF	69.9
	Polyethylene-40 hydrogenated castor oil	Kolliphor® RH40	41.6
	Lauroyl polyoxyl-32 glycerides	Gelucire® 44/14	50.0
Co-surfactants (water-insoluble)	Oleyl polyoxyl-6 glycerides	Labrafil® M1944CS	2.2
	Propylene glycol monocaprylate	Capryol™ 90	14.4
	Propylene glycol monolaurate	Lauroglycol™ 90	4.9
Oils	Medium chain triglycerides	Labrafac™ CC	2.7
	Glyceryl monolinoleate	Maisine™ 35-1	1.6
	Glyceryl mono-oleate	Peceol™	1.8
Hydrophilic cosolvent	Diethylene glycol monoethylether	Transcutol® HP	114.8

Table 2: Dispersion of selected prototype formulations.

SELF	DLS Particle Size (Mean ± sd, nm)		Precipitation?
	With curcumin	Placebo	
100% Tween® 80	28 ± 27	10 ± 2	No
100% Gelucire® 48/16	10 ± 0	7 ± 1	No
100% Gelucire® 44/14	18 ± 6	15 ± 2	No
100% Kolliphor® RH40	32 ± 21	9 ± 2	No

• Lipolysis test

Curcumin reacts with the lipase inhibitor classically used during lipolysis testing (formation of a red-color complex between curcumin and boric acids, [4]). Since prototype formulations contain only water-soluble surfactants, no oily phase could be formed during the lipolysis test. Hence, filtration can be a good alternative for the separation of the pellets phase and the aqueous micellar phase. Each lipolysis sample is then immediately filtered on 0.45µm syringe-filter before dilution in methanol. The in vitro digestion test was able to discriminate prototype formulations (Figure 1). Semi-solid and solid formulations take more time to hydrate and deliver curcumin in the micellar phase in comparison to the liquid one. However, after the 10-minute dispersion phase all formulations are able to release 73 to 80% of the dose in the aqueous phase. The effect of lipolysis can be observed for formulations containing Tween® 80, Kolliphor® RH40 and Gelucire® 44/14. A slight decrease of the concentration of curcumin in the micellar phase is observed after 5 minutes and continues for 1h. Gelucire® 48/16 is the only excipient able to maintain more than 80% of the drug in solution during 1h. The calculation of the SRM of these prototype formulations was not as predictive as expected, even if Gelucire® 48/16 led to one of the lowest SRM (0.8) in comparison to other surfactants (Kolliphor® RH40 - SRM=0.8, Tween® - SRM=1.6, and Gelucire® 44/14 - SRM=3.6).

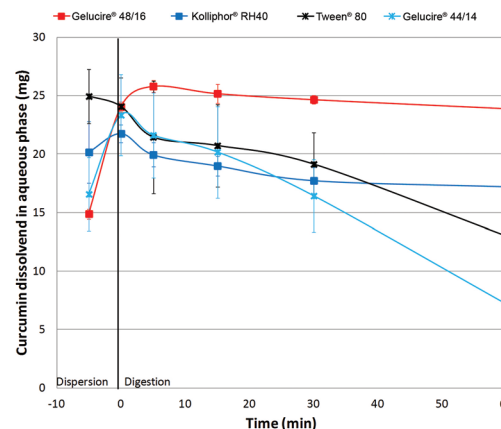


Figure 1: Lipolysis of prototype formulations containing curcumin.

CONCLUSION

LFCS Type IV formulation with Gelucire® 48/16 can efficiently address the low water-solubility of curcumin. pH-stat testing is a valuable tool to discriminate formulations in addition to classic dispersion test.

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